

Supermarket Transactions Analysis

Retail Analytics | Tools: Power BI, DAX, Power Query

Objective

Analyze large-scale retail transaction data to uncover sales trends, category performance and customer purchasing patterns.

Business Problem

Retailers generate huge volumes of transaction-level data but often only look at top-line revenue - missing patterns at the category, basket and time-of-day level that could inform merchandising and staffing decisions.

Dataset

A synthetic supermarket transactions dataset of 55,000+ rows covering products, categories, quantities, prices, timestamps and store locations.

Data Preparation

- Designed and generated a 55,000+ row synthetic transactions dataset
- Built a date/time dimension for hourly and seasonal trend analysis
- Modeled relationships between products, categories and transactions

Analysis Performed

- Revenue and basket-size trends over time
- Category and product-level performance ranking
- Peak sales hours and days by store location
- Basket composition and cross-category purchase patterns

Key KPIs

Total Revenue · Average Basket Size · Units Sold · Top Category Share

Key Insights

- Sales showed clear peak hours that varied by store location.
- A concentrated set of categories drove most of total revenue.
- Basket size and composition varied meaningfully between weekday and weekend transactions.

Recommendations

- Align staffing schedules with identified peak sales hours per location.
- Prioritize shelf space and promotions for top-revenue categories.
- Design weekend-specific bundle promotions based on basket composition patterns.

Skills Demonstrated

DAX, Power Query, Large Dataset Handling, Retail Analytics, Time-based Trend Analysis

